DEMAND FORECASTING AND INVENTORY OPTIMIZATION IN THE FOOD MANUFACTURING INDUSTRY - A FOCUS ON DAIRY PRODUCTS

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[Portfolio role focus: Junior Data Analytics]

1. Business Problem:

Dairy products are highly perishable because these products contain high moisture and nutrients that micro-organisms easily grow on, and it requires efficient inventory and distribution management. Dairy products expire much faster than other food products such as grains, canned food, dry fruits and so on. Poor demand alignment and storage inefficiencies often lead to:

- Product wastage due to expiry
- Revenue loss from inefficient pricing or distribution
- Overstocking or stock shortages
- Inefficient channel allocation

This project analyzes dairy product performance using demand indicators, shelf-life characteristics, and sales channel behavior to identify data-driven strategies for inventory optimization and distribution planning.
The analysis focuses on four key operational metrics:

- Revenue Realization
- Sell-Through Rate
- Average Shelf Life
- Wastage Rate

These insights help businesses improve inventory decisions, storage strategies, and distribution efficiency.

2. Data Overview:

Dataset: Dairy supply chain operational dataset
Website: Opendatabay.com
License: CC0 (Public Domain)
Timeline: 2019 – 2022
Structure of the dataset: 23 features and 4325 instances
Key attributes analyzed:
Product Information

- Product Name
- Brand
- Storage Condition

Sales & Distribution

- Sales Channel (Retail, Wholesale, Online)
- Customer Location

Inventory Data

- Production Date
- Expiration Date
- Quantity Produced
- Quantity Sold
- Quantity in Stock

Operational Metrics

- Shelf Life
- Revenue
- Reorder Quantity
- Minimum Stock Threshold

The dataset enables analysis of demand patterns, product lifecycle behavior, and operational efficiency across multiple dimensions.

3. Key Analytical Insights:

3.1 Revenue Realization: -

Revenue realization measures how efficiently the produced inventory converts into actual revenue.

Key Findings

- Butter and Curd consistently show higher revenue realization compared to other products.

- Some brands such as Amul and Mother Dairy achieve stronger revenue performance across multiple product categories.
- Single-product brands (e.g., Parag Milk Foods – Butter, Passion Cheese – Cheese) show focused specialization but mixed revenue efficiency.

Channel Insights

- Online channels perform better for products like Buttermilk and Yogurt.
- Retail and Wholesale channels show lower revenue realization in several products, suggesting pricing or distribution inefficiencies.

Location Patterns

- Revenue realization varies across customer regions:
- Telangana shows weaker realization in Butter and Ghee.
- Bihar shows weaker realization in Buttermilk and Curd.
- Karnataka underperforms in Ice Cream and Paneer.

These patterns indicate regional demand variation and possible pricing mismatches.

3.2. Sell-Through Performance: -

Sell-through rate measures how effectively produced inventory is sold.

Key Findings

- Most products maintain sell-through rates between 48%–55%, indicating moderate demand stability.
- Amul and Mother Dairy maintain consistent sell-through performance across several products.
- Single-product brands demonstrate very high sell-through efficiency in their specialized products.

Channel Performance

- Online and Retail channels demonstrate stronger sell-through performance than Wholesale.
- Wholesale channels show stable but slightly lower conversion efficiency.

Storage Insights

- Different storage conditions influence product demand behavior:
- Tetra Pack milk achieves the highest sell-through efficiency.
- Refrigerated Curd shows comparatively weaker sales performance.

This suggests shelf-life constraints may influence sales conversion.

3.3. Shelf-Life Analysis: -

Shelf life strongly influences inventory risk and wastage potential.

3.3.1. Product Shelf-Life Categories are given below in table format:

Sl.No	*Shelf-Life Category*	Products
1	Short Shelf Life (<15 days) -These products require fast inventory turnover.	Curd (~6 days) Buttermilk(~11days) Paneer (~11 days) Milk (~13–14 days) Lassi (~14–15 days)
2	Medium Shelf Life (~1 month)	Yogurt Ice Cream Butter
3	Long Shelf Life (>1 month)	Cheese (~56 days) Ghee (~105 days)

3.3.2. Key Insight

- Ghee shows the longest shelf life and zero wastage, making it operationally low risk.
- Curd shows the shortest shelf life, making it highly sensitive to inventory mismanagement.

3.4. Wastage Risk Analysis

Wastage represents unsold inventory that expires before sale.

Product Risk

- Curd shows the highest wastage rates, largely due to its extremely short shelf life.
- Ghee shows zero wastage, confirming its operational advantage.
- Milk packed in Polythene packets shows significantly higher wastage compared to Tetra Pack packaging.

Channel Insights

- Wholesale channels generate the highest wastage for several products.
- Retail channels generally produce lower wastage rates.

Regional Demand

Certain regions demonstrate strong demand signals:

- Karnataka shows low wastage for Ice Cream and Paneer.
- Delhi shows extremely low wastage for the Cheese, indicating strong local demand.

4. Operational Recommendations

Based on the analysis, several strategic improvements are recommended.

4.1. Inventory Strategy

Reduce reorder levels for products with:

1. Sell-through below 45%
2. High wastage risk
3. Short shelf life

4.2 Distribution Strategy

Increase allocation to channels showing strong demand:

1. Online channel for Yogurt and Buttermilk
2. Retail channel for Cheese and Ice Cream

4.3 Storage Optimization

Prioritize improved storage management for:

1. Curd
2. Buttermilk
3. Paneer

These products require faster turnover and tighter inventory monitoring.

4.4. Packaging Improvements

Shift Milk distribution from Polythene packets to Tetra Pack, which demonstrates significantly lower wastage rates.

5. Tools & Technologies

This project integrates multiple analytical tools:

1	SQL (SSMS)	Data cleaning Business rule modeling KPI calculations
2	Tableau	Exploratory dashboard analytics Interactive dimension analysis
3	Power BI	Operational decision dashboards KPI monitoring
4	DAX & Business Rules	Inventory optimization modelling Reorder adjustment calculations